THE ORIGINS OF BITCOIN

MODULE 1.3 EVIDENCE LAB

Problem. Components. Claims. Boundaries.

COGNIZEN CONSULTING

01 THE CONFLICT TRACE

Why a signature is not enough

Trace two transactions that try to spend the same input. The goal is to identify the exact decision the network must make.

EVENT	WHAT CAN BE VERIFIED?	WHAT REMAINS UNRESOLVED?	RULE NEEDED
Alice signs payment to Bob			
Alice signs conflicting payment to Carol			
Bob sees his transaction first			
Carol sees hers first			
A block includes one transaction			
A later block extends that history			

One-sentence design problem

02 THE ANCESTRY MAP

A system, not one magic ingredient

Map each component to its direct job. Then explain how the components reinforce one another.

COMPONENT	DIRECT JOB	WHAT IT CANNOT DO ALONE	BITCOIN COMBINATION
Digital signature			
Cryptographic hash			
Merkle tree			
Timestamp chain			
Proof of work			
Peer-to-peer network			
Node validation			
Economic incentive			

The system-level breakthrough was...

03 THE CLAIM AUDIT

Separate protocol from narrative

Write the claim exactly. Record the mechanism and assumption before deciding what later evidence is still required.

CLAIM	SOURCE / SECTION	MECHANISM	ASSUMPTION	LATER EVIDENCE
Peer-to-peer transfer				
Double-spend resistance				
History becomes costly to replace				
Incentives support participation				
Limited issuance				
Future market value				

Which claim requires the largest inferential leap? Why?

04 CLOSE READING

Read the white paper with boundaries

Use the primary document first. Quote sparingly and record the section so another reader can check your interpretation.

The paper's stated problem is...

The proposed mechanism is...

The most important security assumption is...

A real-world risk outside the paper's mechanism is...

A market claim that requires separate evidence is...

SOURCE CHECK	**Primary source: Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System. Technical context: NISTIR 8202, Blockchain Technology Overview.**

05 PROOF OF LEARNING

Defend a bounded conclusion

Prepare a two-minute explanation for a sceptical colleague. Your conclusion must remain useful without exaggeration.

1. Explain the double-spend problem in plain language

2. Name four contributing technologies and their jobs

3. State one white-paper claim and one non-claim

4. Give one surviving operational risk

5. Write your final bounded conclusion

KEEP THIS

Bitcoin's historical importance is best understood as a coordinated system for ledger agreement—not as proof of every later story told about it.